

Quantitative Analysis of Intrahour Price Boundary Formation: NQ1

Abstract: This study examines the probabilistic distribution of daily High and Low price formation across sixty-minute temporal clusters, specifically focusing on the first fifteen-minute interval (Q1). The objective is to determine the statistical significance of initial momentum as a range-boundary predictor.

I. Methodology & Data Infrastructure

1. Data Source and Resolution

- **Instrument:** NQ1 (Continuous Futures Contract)
- **Resolution:** 1-Minute OHLC Intervals
- **Temporal Scope:** 2006 - 2026
- **Exclusion Criteria:** Partial sessions (hours with fewer than 4 quarters) and non-trading holidays excluded.

2. Time-Series Alignment

- **Primary Timezone:** America/New_York (EST/EDT)
- **Preprocessing:** Source UTC timestamps localized to Exchange Time to account for RTH/Globex session boundaries.
- **Observation Window:** 18:00 (Session Open) to 16:00 (Session Close).

3. Statistical Categorization (Quarters)

The 60-minute hour is discretized into four equal 15-minute segments:

- **Q1 (Initial Quarter):** Minutes :00 - :14
- **Q2:** Minutes :15 - :29
- **Q3:** Minutes :30 - :44
- **Q4:** Minutes :45 - :59

4. Mutual Exclusivity Logic for Q1 Influence

To ensure statistical integrity, Table 3 utilizes a mutually exclusive classification system:

- **Q1 High Only:** Q1 interval contains the final hourly High; Low is established later.
- **Q1 Low Only:** Q1 interval contains the final hourly Low; High is established later.
- **Q1 Both:** Both High and Low are established within the Q1 window (Range Contained).
- **Neither (Expansion):** Both High and Low are broken later in the hour (Range Expansion).

5. Q1-Q4 Opposite Extremes Definition

A **Q1-Q4 Trend Reversal** occurs when Q1 establishes one boundary and Q4 establishes the opposite:

- **Q1 High / Q4 Low:** Price peaks early, declines through the hour (bearish trend).
- **Q1 Low / Q4 High:** Price troughs early, rallies through the hour (bullish trend).

These represent the cleanest directional hours with maximum trend duration.

II. Statistical Results

1. High of Hour Probability

Hour	Total	Q1 High	Q2 High	Q3 High	Q4 High
18:00	4981	2824 (56.7%)	685 (13.8%)	621 (12.5%)	851 (17.1%)
19:00	4969	1905 (38.3%)	785 (15.8%)	841 (16.9%)	1438 (28.9%)
20:00	4905	2031 (41.4%)	820 (16.7%)	802 (16.4%)	1252 (25.5%)
21:00	4900	1801 (36.8%)	902 (18.4%)	911 (18.6%)	1286 (26.2%)
22:00	4877	1984 (40.7%)	862 (17.7%)	895 (18.4%)	1136 (23.3%)
23:00	4849	1872 (38.6%)	900 (18.6%)	894 (18.4%)	1183 (24.4%)
00:00	4917	1843 (37.5%)	917 (18.6%)	896 (18.2%)	1261 (25.6%)
01:00	5004	1830 (36.6%)	809 (16.2%)	887 (17.7%)	1478 (29.5%)
02:00	5094	2056 (40.4%)	771 (15.1%)	888 (17.4%)	1379 (27.1%)
03:00	5121	2118 (41.4%)	885 (17.3%)	827 (16.1%)	1291 (25.2%)

Hour	Total	Q1 High	Q2 High	Q3 High	Q4 High
04:00	5126	2113 (41.2%)	853 (16.6%)	842 (16.4%)	1318 (25.7%)
05:00	5121	1885 (36.8%)	901 (17.6%)	904 (17.7%)	1431 (27.9%)
06:00	5136	1946 (37.9%)	832 (16.2%)	911 (17.7%)	1447 (28.2%)
07:00	5135	1872 (36.5%)	854 (16.6%)	957 (18.6%)	1452 (28.3%)
08:00	5145	1620 (31.5%)	852 (16.6%)	1365 (26.5%)	1308 (25.4%)
09:00	5138	852 (16.6%)	428 (8.3%)	1828 (35.6%)	2030 (39.5%)
10:00	5146	2019 (39.2%)	804 (15.6%)	850 (16.5%)	1473 (28.6%)
11:00	5101	1793 (35.1%)	894 (17.5%)	901 (17.7%)	1513 (29.7%)
12:00	5101	1820 (35.7%)	849 (16.6%)	904 (17.7%)	1528 (30.0%)
13:00	4981	1738 (34.9%)	804 (16.1%)	891 (17.9%)	1548 (31.1%)
14:00	4979	1680 (33.7%)	792 (15.9%)	888 (17.8%)	1619 (32.5%)
15:00	4970	1500 (30.2%)	763 (15.4%)	821 (16.5%)	1886 (37.9%)
16:00	1867	1187 (63.6%)	135 (7.2%)	224 (12.0%)	321 (17.2%)

2. Low of Hour Probability

Hour	Total	Q1 Low	Q2 Low	Q3 Low	Q4 Low
18:00	4981	2890 (58.0%)	680 (13.7%)	590 (11.8%)	821 (16.5%)
19:00	4969	2076 (41.8%)	853 (17.2%)	765 (15.4%)	1275 (25.7%)
20:00	4905	2136 (43.5%)	825 (16.8%)	847 (17.3%)	1097 (22.4%)
21:00	4900	1980 (40.4%)	823 (16.8%)	922 (18.8%)	1175 (24.0%)
22:00	4877	2068 (42.4%)	849 (17.4%)	845 (17.3%)	1115 (22.9%)
23:00	4849	2063 (42.5%)	821 (16.9%)	894 (18.4%)	1071 (22.1%)
00:00	4917	1976 (40.2%)	935 (19.0%)	871 (17.7%)	1135 (23.1%)
01:00	5004	2034 (40.6%)	856 (17.1%)	853 (17.0%)	1261 (25.2%)

Hour	Total	Q1 Low	Q2 Low	Q3 Low	Q4 Low
02:00	5094	2327 (45.7%)	782 (15.4%)	757 (14.9%)	1228 (24.1%)
03:00	5121	2115 (41.3%)	962 (18.8%)	832 (16.2%)	1212 (23.7%)
04:00	5126	2252 (43.9%)	895 (17.5%)	768 (15.0%)	1211 (23.6%)
05:00	5121	2064 (40.3%)	880 (17.2%)	858 (16.8%)	1319 (25.8%)
06:00	5136	2059 (40.1%)	901 (17.5%)	871 (17.0%)	1305 (25.4%)
07:00	5135	2138 (41.6%)	819 (15.9%)	830 (16.2%)	1348 (26.3%)
08:00	5145	1630 (31.7%)	859 (16.7%)	1303 (25.3%)	1353 (26.3%)
09:00	5138	809 (15.7%)	381 (7.4%)	1911 (37.2%)	2037 (39.6%)
10:00	5146	2264 (44.0%)	824 (16.0%)	776 (15.1%)	1282 (24.9%)
11:00	5101	2135 (41.9%)	875 (17.2%)	818 (16.0%)	1273 (25.0%)
12:00	5101	2070 (40.6%)	889 (17.4%)	831 (16.3%)	1311 (25.7%)
13:00	4981	1950 (39.1%)	861 (17.3%)	824 (16.5%)	1346 (27.0%)
14:00	4979	2029 (40.8%)	845 (17.0%)	820 (16.5%)	1285 (25.8%)
15:00	4970	1686 (33.9%)	755 (15.2%)	695 (14.0%)	1834 (36.9%)
16:00	1867	1284 (68.8%)	139 (7.4%)	190 (10.2%)	254 (13.6%)

3. Q1 Influence & Range Expansion

Mutually exclusive breakdown of how Q1 (00-15) affects the hour's range.

Hour	Total	Q1 High Only	Q1 Low Only	Q1 Both	Neither (Exp)
18:00	4981	1702 (34.2%)	1768 (35.5%)	1122 (22.5%)	389 (7.8%)
19:00	4969	1627 (32.7%)	1798 (36.2%)	278 (5.6%)	1266 (25.5%)
20:00	4905	1716 (35.0%)	1821 (37.1%)	315 (6.4%)	1053 (21.5%)
21:00	4900	1569 (32.0%)	1748 (35.7%)	232 (4.7%)	1351 (27.6%)
22:00	4877	1679 (34.4%)	1763 (36.1%)	305 (6.3%)	1130 (23.2%)

Hour	Total	Q1 High Only	Q1 Low Only	Q1 Both	Neither (Exp)
23:00	4849	1590 (32.8%)	1781 (36.7%)	282 (5.8%)	1196 (24.7%)
00:00	4917	1599 (32.5%)	1732 (35.2%)	244 (5.0%)	1342 (27.3%)
01:00	5004	1603 (32.0%)	1807 (36.1%)	227 (4.5%)	1367 (27.3%)
02:00	5094	1722 (33.8%)	1993 (39.1%)	334 (6.6%)	1045 (20.5%)
03:00	5121	1883 (36.8%)	1880 (36.7%)	235 (4.6%)	1123 (21.9%)
04:00	5126	1816 (35.4%)	1955 (38.1%)	297 (5.8%)	1058 (20.6%)
05:00	5121	1683 (32.9%)	1862 (36.4%)	202 (3.9%)	1374 (26.8%)
06:00	5136	1726 (33.6%)	1839 (35.8%)	220 (4.3%)	1351 (26.3%)
07:00	5135	1701 (33.1%)	1967 (38.3%)	171 (3.3%)	1296 (25.2%)
08:00	5145	1534 (29.8%)	1544 (30.0%)	86 (1.7%)	1981 (38.5%)
09:00	5138	844 (16.4%)	801 (15.6%)	8 (0.2%)	3485 (67.8%)
10:00	5146	1828 (35.5%)	2073 (40.3%)	191 (3.7%)	1054 (20.5%)
11:00	5101	1674 (32.8%)	2016 (39.5%)	119 (2.3%)	1292 (25.3%)
12:00	5101	1670 (32.7%)	1920 (37.6%)	150 (2.9%)	1361 (26.7%)
13:00	4981	1624 (32.6%)	1836 (36.9%)	114 (2.3%)	1407 (28.2%)
14:00	4979	1562 (31.4%)	1911 (38.4%)	118 (2.4%)	1388 (27.9%)
15:00	4970	1434 (28.9%)	1620 (32.6%)	66 (1.3%)	1850 (37.2%)
16:00	1867	509 (27.3%)	606 (32.5%)	678 (36.3%)	74 (4.0%)

4. Day of Week Analysis (Q1 Influence Probability)

Cross-sectional variance of Q1 boundary formation (Sun-Fri).

Hour	Sun	Mon	Tue	Wed	Thu	Fri
18:00	97.3%	90.5%	91.3%	91.4%	90.9%	0.0%
19:00	76.0%	75.0%	74.2%	74.4%	73.2%	0.0%

Hour	Sun	Mon	Tue	Wed	Thu	Fri
20:00	77.1%	77.8%	79.4%	78.6%	79.6%	0.0%
21:00	71.8%	71.5%	72.5%	73.3%	73.0%	0.0%
22:00	76.1%	76.9%	76.9%	77.4%	76.9%	0.0%
23:00	74.4%	74.0%	76.2%	74.8%	77.3%	0.0%
00:00	0.0%	72.4%	73.2%	71.3%	72.3%	74.4%
01:00	0.0%	72.1%	75.6%	70.9%	72.1%	72.7%
02:00	0.0%	80.2%	80.5%	80.9%	77.7%	78.0%
03:00	0.0%	78.4%	77.4%	77.6%	77.0%	80.0%
04:00	0.0%	78.7%	79.6%	80.1%	79.1%	79.2%
05:00	0.0%	72.8%	76.3%	70.9%	74.4%	71.5%
06:00	0.0%	74.3%	74.1%	73.7%	72.9%	73.6%
07:00	0.0%	76.3%	74.9%	73.8%	73.0%	75.8%
08:00	0.0%	70.2%	66.0%	62.3%	55.2%	53.7%
09:00	0.0%	29.3%	30.9%	33.0%	31.5%	36.2%
10:00	0.0%	78.7%	78.6%	79.1%	79.7%	81.6%
11:00	0.0%	73.6%	71.3%	77.5%	75.5%	75.4%
12:00	0.0%	71.3%	73.6%	72.8%	75.2%	73.6%
13:00	0.0%	70.6%	71.7%	72.8%	72.1%	71.4%
14:00	0.0%	71.8%	67.9%	74.3%	74.4%	72.2%
15:00	0.0%	60.5%	60.2%	66.5%	64.9%	61.4%
16:00	0.0%	96.4%	96.7%	93.4%	96.1%	98.5%

5. Micro-Temporal Analysis: First Minute Dominance

Probability that the boundary (H or L) for a 15m quarter is formed in its **first minute**.

Hour	Q1 (H/L)	Q2 (H/L)	Q3 (H/L)	Q4 (H/L)
18:00	45.2%/43.6%	21.5%/21.3%	23.7%/22.4%	20.3%/20.6%
19:00	24.0%/23.6%	19.7%/21.1%	21.3%/23.1%	20.7%/21.5%
20:00	27.1%/24.6%	22.4%/23.1%	22.1%/21.1%	21.0%/22.3%
21:00	22.8%/23.0%	21.9%/23.1%	22.6%/22.3%	20.7%/22.1%
22:00	23.2%/21.7%	19.3%/20.1%	21.7%/23.1%	19.7%/21.3%
23:00	21.2%/22.4%	20.2%/20.4%	23.5%/23.0%	19.4%/20.1%
00:00	21.8%/21.7%	18.8%/18.8%	22.9%/22.9%	17.9%/19.1%
01:00	23.5%/23.6%	19.4%/21.1%	21.0%/22.1%	18.2%/18.8%
02:00	28.0%/29.5%	20.5%/21.5%	22.2%/23.8%	19.1%/20.1%
03:00	25.1%/24.0%	21.8%/24.5%	22.4%/23.0%	20.4%/22.0%
04:00	23.9%/26.2%	20.7%/22.3%	21.6%/22.7%	20.0%/21.8%
05:00	22.7%/24.0%	19.6%/21.9%	21.4%/23.0%	20.6%/20.5%
06:00	23.6%/22.4%	20.1%/21.3%	21.2%/22.4%	19.4%/21.8%
07:00	23.7%/27.8%	22.5%/22.2%	22.6%/23.1%	21.0%/21.2%
08:00	23.9%/24.4%	20.2%/21.7%	31.0%/31.4%	22.4%/21.3%
09:00	23.8%/24.5%	17.7%/17.7%	22.9%/24.3%	22.0%/22.4%
10:00	23.8%/23.8%	19.3%/22.6%	21.2%/22.2%	20.8%/20.8%
11:00	20.5%/23.9%	19.7%/21.6%	22.0%/23.7%	19.5%/21.5%
12:00	22.2%/22.8%	20.8%/20.9%	21.0%/21.9%	19.9%/20.4%
13:00	21.8%/21.4%	20.8%/21.5%	21.6%/21.4%	19.7%/21.4%
14:00	22.9%/22.5%	19.2%/21.6%	19.2%/24.2%	19.1%/20.6%
15:00	22.7%/21.9%	19.0%/19.9%	20.4%/23.2%	20.0%/18.5%
16:00	33.7%/38.6%	21.5%/23.2%	31.8%/32.5%	21.7%/23.2%

6. Q1 Breakout Dynamics

Percentage of sessions where the Q1 (00-14) High/Low is violated by subsequent quarters.

Hour	Q1 High Broken By (Q2	Q3	Q4	Never)	Q1 Low Broken By (Q2	Q3	Q4	Never)
18:00	27.6%	10.0%	5.7%	56.7%	27.0%	9.2%	5.8%	58.0%
19:00	39.6%	13.2%	8.8%	38.3%	37.4%	12.8%	7.9%	41.8%
20:00	39.8%	12.0%	6.7%	41.4%	38.9%	11.8%	5.7%	43.5%
21:00	43.9%	13.5%	5.9%	36.8%	40.2%	12.8%	6.5%	40.4%
22:00	40.6%	13.0%	5.7%	40.7%	38.2%	12.8%	6.6%	42.4%
23:00	42.0%	13.1%	6.2%	38.6%	39.0%	12.4%	6.1%	42.5%
00:00	41.5%	14.1%	7.0%	37.5%	40.6%	12.9%	6.4%	40.2%
01:00	41.6%	13.7%	8.2%	36.6%	39.9%	12.0%	7.5%	40.6%
02:00	39.6%	12.5%	7.6%	40.4%	36.2%	11.1%	7.0%	45.7%
03:00	42.7%	10.8%	5.2%	41.4%	41.9%	11.1%	5.7%	41.3%
04:00	42.2%	10.8%	5.8%	41.2%	40.2%	10.3%	5.5%	43.9%
05:00	44.5%	12.7%	6.0%	36.8%	41.5%	11.3%	6.8%	40.3%
06:00	42.7%	12.1%	7.3%	37.9%	41.4%	12.1%	6.4%	40.1%
07:00	44.4%	12.4%	6.7%	36.5%	40.0%	11.6%	6.8%	41.6%
08:00	47.4%	17.0%	4.1%	31.5%	45.9%	17.6%	4.9%	31.7%
09:00	48.0%	31.1%	4.3%	16.6%	48.9%	31.4%	3.9%	15.7%
10:00	44.4%	11.1%	5.3%	39.2%	40.5%	9.6%	5.9%	44.0%
11:00	48.1%	11.0%	5.7%	35.1%	42.1%	10.7%	5.4%	41.9%
12:00	45.5%	12.3%	6.5%	35.7%	43.1%	10.3%	6.0%	40.6%
13:00	46.3%	12.5%	6.2%	34.9%	43.8%	11.6%	5.5%	39.1%

Hour	Q1 High Broken By (Q2	Q3	Q4	Never)	Q1 Low Broken By (Q2	Q3	Q4	Never)
14:00	47.4%	12.8%	6.1%	33.7%	42.8%	10.8%	5.6%	40.8%
15:00	47.4%	13.5%	8.9%	30.2%	44.6%	12.4%	9.1%	33.9%
16:00	18.3%	11.9%	6.2%	63.6%	15.6%	11.1%	4.6%	68.8%

7. Q1-Q4 Trend Reversal (Opposite Extremes)

Probability that the hour's range is bookended by Q1 and Q4 (e.g., Q1 High and Q4 Low).

Hour	Total	Q1 High / Q4 Low	Q1 Low / Q4 High	Total Joint Prob
18:00	4981	676 (13.6%)	717 (14.4%)	28.0%
19:00	4969	802 (16.1%)	912 (18.4%)	34.5%
20:00	4905	741 (15.1%)	861 (17.6%)	32.7%
21:00	4900	689 (14.1%)	790 (16.1%)	30.2%
22:00	4877	734 (15.1%)	727 (14.9%)	30.0%
23:00	4849	658 (13.6%)	758 (15.6%)	29.2%
00:00	4917	697 (14.2%)	773 (15.7%)	29.9%
01:00	5004	758 (15.1%)	917 (18.3%)	33.5%
02:00	5094	839 (16.5%)	958 (18.8%)	35.3%
03:00	5121	831 (16.2%)	867 (16.9%)	33.2%
04:00	5126	814 (15.9%)	908 (17.7%)	33.6%
05:00	5121	781 (15.3%)	910 (17.8%)	33.0%
06:00	5136	810 (15.8%)	895 (17.4%)	33.2%
07:00	5135	838 (16.3%)	933 (18.2%)	34.5%
08:00	5145	730 (14.2%)	678 (13.2%)	27.4%
09:00	5138	562 (10.9%)	531 (10.3%)	21.3%

Hour	Total	Q1 High / Q4 Low	Q1 Low / Q4 High	Total Joint Prob
10:00	5146	865 (16.8%)	1069 (20.8%)	37.6%
11:00	5101	809 (15.9%)	992 (19.4%)	35.3%
12:00	5101	802 (15.7%)	985 (19.3%)	35.0%
13:00	4981	825 (16.6%)	963 (19.3%)	35.9%
14:00	4979	765 (15.4%)	1045 (21.0%)	36.4%
15:00	4970	921 (18.5%)	1027 (20.7%)	39.2%
16:00	1867	231 (12.4%)	293 (15.7%)	28.1%

8. Q1-Q4 Trend Reversal by Day of Week

Total joint probability (Q1 High/Q4 Low + Q1 Low/Q4 High) by day of week.

Hour	Sun	Mon	Tue	Wed	Thu	Fri
18:00	26.0%	29.4%	27.5%	28.3%	28.5%	0.0%
19:00	37.0%	35.1%	33.4%	33.8%	33.4%	0.0%
20:00	32.7%	34.0%	31.8%	33.2%	31.6%	0.0%
21:00	33.3%	27.9%	28.7%	30.8%	30.6%	0.0%
22:00	28.0%	30.4%	30.1%	32.6%	28.6%	0.0%
23:00	32.0%	28.4%	27.5%	29.6%	28.7%	0.0%
00:00	0.0%	28.1%	29.9%	27.9%	29.9%	33.7%
01:00	0.0%	33.5%	34.6%	31.7%	34.0%	33.5%
02:00	0.0%	33.1%	34.3%	35.0%	37.2%	36.8%
03:00	0.0%	32.6%	35.9%	32.6%	32.7%	31.9%
04:00	0.0%	33.1%	33.8%	32.1%	35.2%	33.8%
05:00	0.0%	32.4%	33.4%	31.2%	34.4%	33.7%
06:00	0.0%	33.1%	33.1%	35.4%	32.6%	31.9%

Hour	Sun	Mon	Tue	Wed	Thu	Fri
07:00	0.0%	34.8%	33.7%	32.3%	33.6%	38.1%
08:00	0.0%	31.3%	29.6%	25.9%	25.5%	24.4%
09:00	0.0%	19.4%	21.2%	22.1%	20.4%	23.4%
10:00	0.0%	37.0%	39.0%	37.9%	36.9%	37.1%
11:00	0.0%	35.1%	33.6%	36.5%	36.4%	34.9%
12:00	0.0%	34.8%	34.2%	34.0%	36.8%	35.3%
13:00	0.0%	37.8%	36.4%	35.3%	35.8%	34.2%
14:00	0.0%	36.6%	32.9%	37.5%	39.9%	34.8%
15:00	0.0%	40.3%	37.0%	40.9%	38.5%	39.3%
16:00	0.0%	24.7%	29.8%	28.5%	27.5%	30.3%

8b. Q1-Q4 Detailed DOW Breakdown

Per-day split: Q1H/Q4L (bearish trend) vs Q1L/Q4H (bullish trend).

Hour	Day	N	Q1H/Q4L	Q1L/Q4H	Total
09:00	Mon	1026	10.4%	9.0%	19.4%
09:00	Tue	1034	10.3%	10.9%	21.2%
09:00	Wed	1033	10.5%	11.6%	22.1%
09:00	Thu	1035	10.6%	9.8%	20.4%
09:00	Fri	1010	13.0%	10.4%	23.4%
10:00	Mon	1026	16.8%	20.3%	37.0%
10:00	Tue	1036	17.3%	21.7%	39.0%
10:00	Wed	1037	16.9%	21.0%	37.9%
10:00	Thu	1035	15.5%	21.4%	36.9%
10:00	Fri	1012	17.7%	19.4%	37.1%

Hour	Day	N	Q1H/Q4L	Q1L/Q4H	Total
11:00	Mon	993	14.9%	20.2%	35.1%
11:00	Tue	1036	14.3%	19.3%	33.6%
11:00	Wed	1035	16.1%	20.4%	36.5%
11:00	Thu	1027	16.7%	19.8%	36.4%
11:00	Fri	1010	17.3%	17.5%	34.9%
12:00	Mon	994	14.4%	20.4%	34.8%
12:00	Tue	1036	15.8%	18.3%	34.2%
12:00	Wed	1035	15.6%	18.5%	34.0%
12:00	Thu	1026	17.0%	19.9%	36.8%
12:00	Fri	1010	15.8%	19.5%	35.3%
13:00	Mon	936	17.0%	20.8%	37.8%
13:00	Tue	1027	16.1%	20.4%	36.4%
13:00	Wed	1027	16.0%	19.4%	35.3%
13:00	Thu	1005	17.1%	18.7%	35.8%
13:00	Fri	986	16.7%	17.4%	34.2%
14:00	Mon	936	14.7%	21.9%	36.6%
14:00	Tue	1026	13.3%	19.7%	32.9%
14:00	Wed	1026	16.7%	20.9%	37.5%
14:00	Thu	1005	18.1%	21.8%	39.9%
14:00	Fri	986	14.0%	20.8%	34.8%
15:00	Mon	935	18.7%	21.6%	40.3%
15:00	Tue	1026	18.7%	18.3%	37.0%
15:00	Wed	1025	19.7%	21.2%	40.9%
15:00	Thu	1004	18.5%	20.0%	38.5%

Hour	Day	N	Q1H/Q4L	Q1L/Q4H	Total
15:00	Fri	980	16.9%	22.3%	39.3%

9. Dominant Quarter per Hour

Which quarter most frequently sets the High and Low of each hour. **Bold** = dominant quarter.

Hour	High→Q	High%	Low→Q	Low%	Q2+Q3 High%	Q2+Q3 Low%
18:00	Q1	56.7%	Q1	58.0%	26.2%	25.5%
19:00	Q1	38.3%	Q1	41.8%	32.7%	32.6%
20:00	Q1	41.4%	Q1	43.5%	33.1%	34.1%
21:00	Q1	36.8%	Q1	40.4%	37.0%	35.6%
22:00	Q1	40.7%	Q1	42.4%	36.0%	34.7%
23:00	Q1	38.6%	Q1	42.5%	37.0%	35.4%
00:00	Q1	37.5%	Q1	40.2%	36.9%	36.7%
01:00	Q1	36.6%	Q1	40.6%	33.9%	34.2%
02:00	Q1	40.4%	Q1	45.7%	32.6%	30.2%
03:00	Q1	41.4%	Q1	41.3%	33.4%	35.0%
04:00	Q1	41.2%	Q1	43.9%	33.1%	32.4%
05:00	Q1	36.8%	Q1	40.3%	35.2%	33.9%
06:00	Q1	37.9%	Q1	40.1%	33.9%	34.5%
07:00	Q1	36.5%	Q1	41.6%	35.3%	32.1%
08:00	Q1	31.5%	Q1	31.7%	43.1%	42.0%
09:00	Q4	39.5%	Q4	39.6%	43.9%	44.6%
10:00	Q1	39.2%	Q1	44.0%	32.1%	31.1%
11:00	Q1	35.1%	Q1	41.9%	35.2%	33.2%
12:00	Q1	35.7%	Q1	40.6%	34.4%	33.7%

Hour	High→Q	High%	Low→Q	Low%	Q2+Q3 High%	Q2+Q3 Low%
13:00	Q1	34.9%	Q1	39.1%	34.0%	33.8%
14:00	Q1	33.7%	Q1	40.8%	33.7%	33.4%
15:00	Q4	37.9%	Q4	36.9%	31.9%	29.2%
16:00	Q1	63.6%	Q1	68.8%	19.2%	17.6%

9b. Middle Quarter (Q2/Q3) Analysis

Breakdown of middle-quarter boundary formation by hour.

Hour	Q2 High%	Q3 High%	Q2 Low%	Q3 Low%	Q2+Q3 Combined	Interpretation
18:00	13.8%	12.5%	13.7%	11.8%	51.7%	Q2 > Q3 — early continuation
19:00	15.8%	16.9%	17.2%	15.4%	65.3%	Balanced mid-quarters
20:00	16.7%	16.4%	16.8%	17.3%	67.2%	Balanced mid-quarters
21:00	18.4%	18.6%	16.8%	18.8%	72.6%	Q3 > Q2 — late-hour reversal zone
22:00	17.7%	18.4%	17.4%	17.3%	70.8%	Balanced mid-quarters
23:00	18.6%	18.4%	16.9%	18.4%	72.4%	Balanced mid-quarters
00:00	18.6%	18.2%	19.0%	17.7%	73.6%	Q2 > Q3 — early continuation
01:00	16.2%	17.7%	17.1%	17.0%	68.0%	Balanced mid-quarters
02:00	15.1%	17.4%	15.4%	14.9%	62.8%	Balanced mid-quarters

Hour	Q2 High%	Q3 High%	Q2 Low%	Q3 Low%	Q2+Q3 Combined	Interpretation
03:00	17.3%	16.1%	18.8%	16.2%	68.5%	Q2 > Q3 — early continuation
04:00	16.6%	16.4%	17.5%	15.0%	65.5%	Q2 > Q3 — early continuation
05:00	17.6%	17.7%	17.2%	16.8%	69.2%	Balanced mid-quarters
06:00	16.2%	17.7%	17.5%	17.0%	68.4%	Balanced mid-quarters
07:00	16.6%	18.6%	15.9%	16.2%	67.4%	Q3 > Q2 — late-hour reversal zone
08:00	16.6%	26.5%	16.7%	25.3%	85.1%	Mid-Q dominant — Q1 unreliable
09:00	8.3%	35.6%	7.4%	37.2%	88.5%	Mid-Q dominant — Q1 unreliable
10:00	15.6%	16.5%	16.0%	15.1%	63.2%	Balanced mid-quarters
11:00	17.5%	17.7%	17.2%	16.0%	68.4%	Balanced mid-quarters
12:00	16.6%	17.7%	17.4%	16.3%	68.1%	Balanced mid-quarters
13:00	16.1%	17.9%	17.3%	16.5%	67.9%	Balanced mid-quarters
14:00	15.9%	17.8%	17.0%	16.5%	67.2%	Balanced mid-quarters
15:00	15.4%	16.5%	15.2%	14.0%	61.0%	Balanced mid-quarters
16:00	7.2%	12.0%	7.4%	10.2%	36.9%	Q3 > Q2 — late-hour reversal zone

10. Joint Distribution: High Quarter x Low Quarter

Full 4x4 matrix showing which quarter makes the High AND which makes the Low. All 16 cells sum to 100%.

18:00 (N=4981)

	Low=Q1	Low=Q2	Low=Q3	Low=Q4	Row
Hi=Q1	22.5%	10.5%	10.1%	13.6%	56.7%
Hi=Q2	10.5%	0.3%	1.0%	1.9%	13.8%
Hi=Q3	10.6%	0.8%	0.1%	0.9%	12.5%
Hi=Q4	14.4%	2.0%	0.6%	0.1%	17.1%
Col	58.0%	13.7%	11.8%	16.5%	100%

19:00 (N=4969)

	Low=Q1	Low=Q2	Low=Q3	Low=Q4	Row
Hi=Q1	5.6%	7.4%	9.2%	16.1%	38.3%
Hi=Q2	7.0%	0.9%	2.2%	5.8%	15.8%
Hi=Q3	10.9%	2.6%	0.7%	2.7%	16.9%
Hi=Q4	18.4%	6.3%	3.3%	1.0%	28.9%
Col	41.8%	17.2%	15.4%	25.7%	100%

20:00 (N=4905)

	Low=Q1	Low=Q2	Low=Q3	Low=Q4	Row
Hi=Q1	6.4%	8.1%	11.8%	15.1%	41.4%
Hi=Q2	8.3%	0.9%	2.8%	4.7%	16.7%
Hi=Q3	11.2%	2.7%	0.5%	2.0%	16.4%
Hi=Q4	17.6%	5.2%	2.3%	0.6%	25.5%
Col	43.5%	16.8%	17.3%	22.4%	100%

21:00 (N=4900)

	Low=Q1	Low=Q2	Low=Q3	Low=Q4	Row
Hi=Q1	4.7%	6.7%	11.3%	14.1%	36.8%
Hi=Q2	7.8%	1.0%	3.4%	6.3%	18.4%
Hi=Q3	11.8%	3.0%	0.8%	3.1%	18.6%
Hi=Q4	16.1%	6.2%	3.4%	0.5%	26.2%
Col	40.4%	16.8%	18.8%	24.0%	100%

22:00 (N=4877)

	Low=Q1	Low=Q2	Low=Q3	Low=Q4	Row
Hi=Q1	6.3%	8.0%	11.4%	15.1%	40.7%
Hi=Q2	8.7%	1.0%	2.7%	5.2%	17.7%
Hi=Q3	12.5%	2.9%	0.7%	2.2%	18.4%
Hi=Q4	14.9%	5.5%	2.5%	0.4%	23.3%
Col	42.4%	17.4%	17.3%	22.9%	100%

23:00 (N=4849)

	Low=Q1	Low=Q2	Low=Q3	Low=Q4	Row
Hi=Q1	5.8%	7.5%	11.7%	13.6%	38.6%
Hi=Q2	8.7%	0.9%	3.4%	5.5%	18.6%
Hi=Q3	12.4%	2.7%	0.9%	2.5%	18.4%
Hi=Q4	15.6%	5.8%	2.5%	0.5%	24.4%
Col	42.5%	16.9%	18.4%	22.1%	100%

00:00 (N=4917)

	Low=Q1	Low=Q2	Low=Q3	Low=Q4	Row
Hi=Q1	5.0%	8.0%	10.4%	14.2%	37.5%
Hi=Q2	8.0%	1.3%	3.5%	5.8%	18.6%
Hi=Q3	11.5%	3.3%	0.9%	2.5%	18.2%
Hi=Q4	15.7%	6.3%	3.0%	0.6%	25.6%
Col	40.2%	19.0%	17.7%	23.1%	100%

01:00 (N=5004)

	Low=Q1	Low=Q2	Low=Q3	Low=Q4	Row
Hi=Q1	4.5%	6.8%	10.1%	15.1%	36.6%
Hi=Q2	6.6%	0.6%	2.9%	6.0%	16.2%
Hi=Q3	11.2%	2.9%	0.6%	3.0%	17.7%
Hi=Q4	18.3%	6.8%	3.4%	1.0%	29.5%
Col	40.6%	17.1%	17.0%	25.2%	100%

02:00 (N=5094)

	Low=Q1	Low=Q2	Low=Q3	Low=Q4	Row
Hi=Q1	6.6%	7.2%	10.2%	16.5%	40.4%
Hi=Q2	8.4%	0.5%	1.7%	4.6%	15.1%
Hi=Q3	11.9%	2.4%	0.5%	2.6%	17.4%
Hi=Q4	18.8%	5.4%	2.5%	0.4%	27.1%
Col	45.7%	15.4%	14.9%	24.1%	100%

03:00 (N=5121)

	Low=Q1	Low=Q2	Low=Q3	Low=Q4	Row
Hi=Q1	4.6%	9.0%	11.5%	16.2%	41.4%

	Low=Q1	Low=Q2	Low=Q3	Low=Q4	Row
Hi=Q2	8.1%	0.7%	2.5%	6.0%	17.3%
Hi=Q3	11.7%	3.0%	0.3%	1.2%	16.1%
Hi=Q4	16.9%	6.1%	2.0%	0.2%	25.2%
Col	41.3%	18.8%	16.2%	23.7%	100%

04:00 (N=5126)

	Low=Q1	Low=Q2	Low=Q3	Low=Q4	Row
Hi=Q1	5.8%	9.1%	10.4%	15.9%	41.2%
Hi=Q2	8.6%	0.6%	2.2%	5.2%	16.6%
Hi=Q3	11.9%	2.2%	0.3%	2.1%	16.4%
Hi=Q4	17.7%	5.4%	2.1%	0.4%	25.7%
Col	43.9%	17.5%	15.0%	23.6%	100%

05:00 (N=5121)

	Low=Q1	Low=Q2	Low=Q3	Low=Q4	Row
Hi=Q1	3.9%	7.2%	10.4%	15.3%	36.8%
Hi=Q2	7.3%	0.9%	2.6%	6.8%	17.6%
Hi=Q3	11.3%	2.8%	0.6%	2.9%	17.7%
Hi=Q4	17.8%	6.4%	3.0%	0.8%	27.9%
Col	40.3%	17.2%	16.8%	25.8%	100%

06:00 (N=5136)

	Low=Q1	Low=Q2	Low=Q3	Low=Q4	Row
Hi=Q1	4.3%	7.3%	10.6%	15.8%	37.9%
Hi=Q2	6.7%	0.7%	2.7%	6.1%	16.2%

	Low=Q1	Low=Q2	Low=Q3	Low=Q4	Row
Hi=Q3	11.7%	2.7%	0.6%	2.7%	17.7%
Hi=Q4	17.4%	6.8%	3.1%	0.9%	28.2%
Col	40.1%	17.5%	17.0%	25.4%	100%

07:00 (N=5135)

	Low=Q1	Low=Q2	Low=Q3	Low=Q4	Row
Hi=Q1	3.3%	6.3%	10.5%	16.3%	36.5%
Hi=Q2	7.7%	0.7%	2.1%	6.1%	16.6%
Hi=Q3	12.5%	2.3%	0.6%	3.2%	18.6%
Hi=Q4	18.2%	6.6%	2.9%	0.6%	28.3%
Col	41.6%	15.9%	16.2%	26.3%	100%

08:00 (N=5145)

	Low=Q1	Low=Q2	Low=Q3	Low=Q4	Row
Hi=Q1	1.7%	4.5%	11.1%	14.2%	31.5%
Hi=Q2	4.8%	0.5%	4.8%	6.4%	16.6%
Hi=Q3	12.0%	4.9%	4.4%	5.2%	26.5%
Hi=Q4	13.2%	6.8%	5.0%	0.4%	25.4%
Col	31.7%	16.7%	25.3%	26.3%	100%

09:00 (N=5138)

	Low=Q1	Low=Q2	Low=Q3	Low=Q4	Row
Hi=Q1	0.2%	0.2%	5.3%	10.9%	16.6%
Hi=Q2	0.2%	0.1%	2.3%	5.8%	8.3%
Hi=Q3	5.1%	2.2%	8.6%	19.8%	35.6%

	Low=Q1	Low=Q2	Low=Q3	Low=Q4	Row
Hi=Q4	10.3%	5.0%	21.0%	3.2%	39.5%
Col	15.7%	7.4%	37.2%	39.6%	100%

10:00 (N=5146)

	Low=Q1	Low=Q2	Low=Q3	Low=Q4	Row
Hi=Q1	3.7%	7.8%	10.9%	16.8%	39.2%
Hi=Q2	7.3%	0.5%	2.0%	5.9%	15.6%
Hi=Q3	12.2%	1.9%	0.3%	2.1%	16.5%
Hi=Q4	20.8%	5.8%	1.9%	0.1%	28.6%
Col	44.0%	16.0%	15.1%	24.9%	100%

11:00 (N=5101)

	Low=Q1	Low=Q2	Low=Q3	Low=Q4	Row
Hi=Q1	2.3%	6.7%	10.2%	15.9%	35.1%
Hi=Q2	7.7%	0.7%	2.7%	6.4%	17.5%
Hi=Q3	12.4%	2.3%	0.5%	2.5%	17.7%
Hi=Q4	19.4%	7.4%	2.6%	0.2%	29.7%
Col	41.9%	17.2%	16.0%	25.0%	100%

12:00 (N=5101)

	Low=Q1	Low=Q2	Low=Q3	Low=Q4	Row
Hi=Q1	2.9%	7.0%	10.0%	15.7%	35.7%
Hi=Q2	6.7%	0.7%	2.8%	6.4%	16.6%
Hi=Q3	11.6%	2.7%	0.5%	3.0%	17.7%
Hi=Q4	19.3%	7.0%	3.0%	0.6%	30.0%

	Low=Q1	Low=Q2	Low=Q3	Low=Q4	Row
Col	40.6%	17.4%	16.3%	25.7%	100%

13:00 (N=4981)

	Low=Q1	Low=Q2	Low=Q3	Low=Q4	Row
Hi=Q1	2.3%	6.0%	10.1%	16.6%	34.9%
Hi=Q2	6.2%	0.6%	2.6%	6.8%	16.1%
Hi=Q3	11.3%	2.6%	0.8%	3.2%	17.9%
Hi=Q4	19.3%	8.2%	3.1%	0.5%	31.1%
Col	39.1%	17.3%	16.5%	27.0%	100%

14:00 (N=4979)

	Low=Q1	Low=Q2	Low=Q3	Low=Q4	Row
Hi=Q1	2.4%	6.0%	10.0%	15.4%	33.7%
Hi=Q2	5.9%	0.7%	2.6%	6.7%	15.9%
Hi=Q3	11.5%	2.5%	0.6%	3.2%	17.8%
Hi=Q4	21.0%	7.7%	3.3%	0.5%	32.5%
Col	40.8%	17.0%	16.5%	25.8%	100%

15:00 (N=4970)

	Low=Q1	Low=Q2	Low=Q3	Low=Q4	Row
Hi=Q1	1.3%	3.7%	6.7%	18.5%	30.2%
Hi=Q2	3.6%	0.3%	1.6%	9.8%	15.4%
Hi=Q3	8.4%	2.4%	0.4%	5.4%	16.5%
Hi=Q4	20.7%	8.8%	5.3%	3.1%	37.9%
Col	33.9%	15.2%	14.0%	36.9%	100%

16:00 (N=1867)

	Low=Q1	Low=Q2	Low=Q3	Low=Q4	Row
Hi=Q1	36.3%	5.8%	9.1%	12.4%	63.6%
Hi=Q2	5.5%	0.6%	0.3%	0.9%	7.2%
Hi=Q3	11.2%	0.2%	0.3%	0.3%	12.0%
Hi=Q4	15.7%	0.9%	0.5%	0.1%	17.2%
Col	68.8%	7.4%	10.2%	13.6%	100%

11. Where Does the Opposite Extreme Land When Q1 Sets One Boundary?

Given Q1 makes the High (or Low), which quarter makes the opposite boundary?

Hour	Q1 Hi Only	...Q4 Low	...Q2/Q3 Low	Q4 Share	Q1 Lo Only	...Q4 High	...Q2/Q3 High	Q4 Share
18:00	34.2%	13.6%	20.6%	40%	35.5%	14.4%	21.1%	41%
19:00	32.7%	16.1%	16.6%	49%	36.2%	18.4%	17.8%	51%
20:00	35.0%	15.1%	19.9%	43%	37.1%	17.6%	19.6%	47%
21:00	32.0%	14.1%	18.0%	44%	35.7%	16.1%	19.6%	45%
22:00	34.4%	15.1%	19.4%	44%	36.1%	14.9%	21.2%	41%
23:00	32.8%	13.6%	19.2%	41%	36.7%	15.6%	21.1%	43%
00:00	32.5%	14.2%	18.3%	44%	35.2%	15.7%	19.5%	45%
01:00	32.0%	15.1%	16.9%	47%	36.1%	18.3%	17.8%	51%
02:00	33.8%	16.5%	17.3%	49%	39.1%	18.8%	20.3%	48%
03:00	36.8%	16.2%	20.5%	44%	36.7%	16.9%	19.8%	46%
04:00	35.4%	15.9%	19.5%	45%	38.1%	17.7%	20.4%	46%
05:00	32.9%	15.3%	17.6%	46%	36.4%	17.8%	18.6%	49%
06:00	33.6%	15.8%	17.8%	47%	35.8%	17.4%	18.4%	49%

Hour	Q1 Hi Only	...Q4 Low	...Q2/Q3 Low	Q4 Share	Q1 Lo Only	...Q4 High	...Q2/Q3 High	Q4 Share
07:00	33.1%	16.3%	16.8%	49%	38.3%	18.2%	20.1%	47%
08:00	29.8%	14.2%	15.6%	48%	30.0%	13.2%	16.8%	44%
09:00	16.4%	10.9%	5.5%	67%	15.6%	10.3%	5.3%	66%
10:00	35.5%	16.8%	18.7%	47%	40.3%	20.8%	19.5%	52%
11:00	32.8%	15.9%	17.0%	48%	39.5%	19.4%	20.1%	49%
12:00	32.7%	15.7%	17.0%	48%	37.6%	19.3%	18.3%	51%
13:00	32.6%	16.6%	16.0%	51%	36.9%	19.3%	17.5%	52%
14:00	31.4%	15.4%	16.0%	49%	38.4%	21.0%	17.4%	55%
15:00	28.9%	18.5%	10.3%	64%	32.6%	20.7%	11.9%	63%
16:00	27.3%	12.4%	14.9%	45%	32.5%	15.7%	16.8%	48%

Reading this table: When Q1 sets the High ('Q1 Hi Only'), the Low can land in Q2, Q3, or Q4. 'Q4 Share' shows what percentage of those Q1-High sessions have Q4 as the Low — the clean trend scenario.

The remainder ('Q2/Q3 Low') represents mid-hour reversals where the opposite extreme is set before Q4.

III. Overall Analysis & Tactical Implications

A. Cross-Hour Summary

Sorted by Q1-Q4 opposite extreme probability (highest first).

Rank	Hour	N	Q1 Sets H or L	Q1H/Q4L	Q1L/Q4H	Q1-Q4 Total	Expansion
1	15:00	4970	62.8%	18.5%	20.7%	39.2%	37.2%
2	10:00	5146	79.5%	16.8%	20.8%	37.6%	20.5%

Rank	Hour	N	Q1 Sets H or L	Q1H/Q4L	Q1L/Q4H	Q1-Q4 Total	Expansion
3	14:00	4979	72.1%	15.4%	21.0%	36.4%	27.9%
4	13:00	4981	71.8%	16.6%	19.3%	35.9%	28.2%
5	11:00	5101	74.7%	15.9%	19.4%	35.3%	25.3%
6	02:00	5094	79.5%	16.5%	18.8%	35.3%	20.5%
7	12:00	5101	73.3%	15.7%	19.3%	35.0%	26.7%
8	19:00	4969	74.5%	16.1%	18.4%	34.5%	25.5%
9	07:00	5135	74.8%	16.3%	18.2%	34.5%	25.2%
10	04:00	5126	79.4%	15.9%	17.7%	33.6%	20.6%
11	01:00	5004	72.7%	15.1%	18.3%	33.5%	27.3%
12	06:00	5136	73.7%	15.8%	17.4%	33.2%	26.3%
13	03:00	5121	78.1%	16.2%	16.9%	33.2%	21.9%
14	05:00	5121	73.2%	15.3%	17.8%	33.0%	26.8%
15	20:00	4905	78.5%	15.1%	17.6%	32.7%	21.5%
16	21:00	4900	72.4%	14.1%	16.1%	30.2%	27.6%
17	22:00	4877	76.8%	15.1%	14.9%	30.0%	23.2%
18	00:00	4917	72.7%	14.2%	15.7%	29.9%	27.3%
19	23:00	4849	75.3%	13.6%	15.6%	29.2%	24.7%
20	16:00	1867	96.0%	12.4%	15.7%	28.1%	4.0%
21	18:00	4981	92.2%	13.6%	14.4%	28.0%	7.8%
22	08:00	5145	61.5%	14.2%	13.2%	27.4%	38.5%
23	09:00	5138	32.2%	10.9%	10.3%	21.3%	67.8%

B. Session-Level Insights

Session	Avg Q1-Q4 Opp%	Best Hour	Worst Hour	Avg Expansion%
Globex Evening (18-20)	31.7%	19:00 (34.5%)	18:00 (28.0%)	18.3%
Asia (21-01)	30.5%	01:00 (33.5%)	23:00 (29.2%)	26.0%
London (02-07)	33.8%	02:00 (35.3%)	05:00 (33.0%)	23.6%
Pre-Market (08)	27.4%	08:00 (27.4%)	08:00 (27.4%)	38.5%
NY Open (09)	21.3%	09:00 (21.3%)	09:00 (21.3%)	67.8%
NY AM (10-12)	36.0%	10:00 (37.6%)	12:00 (35.0%)	24.2%
NY PM/Lunch (13-14)	36.1%	14:00 (36.4%)	13:00 (35.9%)	28.1%
Power Hour (15)	39.2%	15:00 (39.2%)	15:00 (39.2%)	37.2%
Close (16)	28.1%	16:00 (28.1%)	16:00 (28.1%)	4.0%

C. Bullish vs Bearish Trend Bias

Ratio of Q1L/Q4H (bullish) to Q1H/Q4L (bearish). Ratio > 1.0 = bullish bias.

Hour	Q1L→Q4H (Bull)	Q1H→Q4L (Bear)	Bull/Bear Ratio	Bias
18:00	14.4%	13.6%	1.06	🦏 Neutral
19:00	18.4%	16.1%	1.14	🐂 Bullish
20:00	17.6%	15.1%	1.16	🐂 Bullish

Hour	Q1L→Q4H (Bull)	Q1H→Q4L (Bear)	Bull/Bear Ratio	Bias
21:00	16.1%	14.1%	1.15	🐮 Bullish
22:00	14.9%	15.1%	0.99	🐼 Neutral
23:00	15.6%	13.6%	1.15	🐮 Bullish
00:00	15.7%	14.2%	1.11	🐮 Bullish
01:00	18.3%	15.1%	1.21	🐮 Bullish
02:00	18.8%	16.5%	1.14	🐮 Bullish
03:00	16.9%	16.2%	1.04	🐼 Neutral
04:00	17.7%	15.9%	1.12	🐮 Bullish
05:00	17.8%	15.3%	1.17	🐮 Bullish
06:00	17.4%	15.8%	1.10	🐮 Bullish
07:00	18.2%	16.3%	1.11	🐮 Bullish
08:00	13.2%	14.2%	0.93	🐼 Neutral
09:00	10.3%	10.9%	0.94	🐼 Neutral
10:00	20.8%	16.8%	1.24	🐮 Bullish
11:00	19.4%	15.9%	1.23	🐮 Bullish
12:00	19.3%	15.7%	1.23	🐮 Bullish
13:00	19.3%	16.6%	1.17	🐮 Bullish
14:00	21.0%	15.4%	1.37	🐮 Bullish
15:00	20.7%	18.5%	1.12	🐮 Bullish
16:00	15.7%	12.4%	1.27	🐮 Bullish

D. Key Findings

Highest Q1-Q4 Trend Reversal Hours (Best for Directional Plays)

- 15:00** — 39.2% of sessions see Q1 set one extreme, Q4 the opposite. Bearish trend 18.5% / Bullish trend 20.7%.

2. **10:00** — 37.6% of sessions see Q1 set one extreme, Q4 the opposite. Bearish trend 16.8% / Bullish trend 20.8%.
3. **14:00** — 36.4% of sessions see Q1 set one extreme, Q4 the opposite. Bearish trend 15.4% / Bullish trend 21.0%.

Lowest Q1-Q4 Trend Reversal Hours (Choppy / Expansion)

1. **18:00** — Only 28.0% Q1-Q4 bookends. Expansion (Neither) at 7.8%.
 2. **08:00** — Only 27.4% Q1-Q4 bookends. Expansion (Neither) at 38.5%.
 3. **09:00** — Only 21.3% Q1-Q4 bookends. Expansion (Neither) at 67.8%.
- **Session Open (16:00):** Q1 sets at least one boundary 96.0% of the time. This is the 'Judas' hour — the first 15 minutes sets the tone.
 - **NY Open (09:00):** Highest expansion at 67.8% — Q1 boundaries are broken in both directions. Only 21.3% Q1-Q4 opposite extremes. Avoid fading Q1.
 - **Power Hour (15:00):** Highest Q1-Q4 opposite extreme at 39.2%. Strong directional moves: Bull 20.7% / Bear 18.5%.

Day-of-Week Patterns (RTH Hours)

- **09:00** — Best: Friday (23.4%), Worst: Monday (19.4%)
- **10:00** — Best: Tuesday (39.0%), Worst: Thursday (36.9%)
- **15:00** — Best: Wednesday (40.9%), Worst: Tuesday (37.0%)

E. Quarter Dominance Analysis

Q1 dominates 21 of 23 hours for both High and Low formation. The two exceptions reveal critical regime changes:

Q4-Dominant Hours

- **09:00** — High: Q4 (39.5%), Low: Q4 (39.6%). Maximum expansion hour — Q1 boundaries are noise, not signal.
- **15:00** — High: Q4 (37.9%), Low: Q4 (36.9%). The hour trends from open to close. Q1 is the setup, Q4 is the destination.

High Middle-Quarter Hours (Q2+Q3 Combined > 80%)

- **08:00** — Q2+Q3 combined: 85.1%. Q3 alone: High 26.5% / Low 25.3%. The :30 mark (8:30 news) is the true session pivot — ignore Q1.

- **09:00** — Q2+Q3 combined: 88.5%. Q3 alone: High 35.6% / Low 37.2%. Q1 is essentially random. Wait for the :30 rotation to trade.

Tactical Rules by Quarter Regime

Regime	Hours	Rule
Q1 Dominant (>35%)	Most Globex, London, NY AM	Fade Q1 extreme for mean-reversion, or use Q1 boundary as SL for continuation
Q4 Dominant	09:00, 15:00	Do NOT fade Q1 — let the trend develop. Enter on Q2/Q3 pullbacks toward Q4
Mid-Q Dominant (>80%)	08:00, 09:00	Wait for :30 rotation. Q1 boundaries are unreliable. The real range sets after economic data
Q1 Overwhelm (>60%)	16:00, 18:00	The 'Judas Candle' regime. Q1 IS the range. Fade breakout attempts from Q2-Q4